

Heekyung Kim

Data Platform Engineer

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PROFILE

Cloud and data platform engineer translating **cost, scaling, security and operational constraints into measurable architecture decisions**. Took an Apache Iceberg lakehouse on AWS S3, EKS, MSK, Glue and Athena into production, and investigated compute costs and scheduling bottlenecks in Spark on EKS with Spot and Karpenter. Turn operational knowledge into reusable infrastructure as code, guardrails and reliability checks.

Experience

11/2025 - Present

Wrtn Technologies | Data Engineer

- Built the MongoDB → MSK / Kafka Connect → S3 / Iceberg CDC path and migrated approximately 200 tables to production.
- Analyzed and improved Spot capacity, executor sizing and ephemeral-storage constraints in Spark on EKS / Karpenter.
- Established the multiregion governance IaC repository and standardized Terraform / GitHub Actions operations.

07/2024 - 11/2025

Concentrix Korea | Data Platform Engineer

- Built a feasibility PoC for migrating a client's on-premises YARN / Spark platform to AWS using EMR on EKS and Terraform.
- Redesigned an SEO pipeline as raw-first ELT, reducing API-credit costs from ~\$1,800 to ~\$200 per month (-89%).

06/2022 - 07/2024

Tridge | Data Engineer

- Owned the technical implementation of Korean and Chinese trade-data pipelines: ~20 daily crawlers, ELT and SQL marts.

Selected Projects

Production migration to an AWS CDC / Iceberg lakehouse

05/2026 - Present

Problem	Reduce managed-warehouse CDC apply and query costs while preserving the change stream and consistency of ~200 live tables.
Decisions	Rejected the initial cross-cloud design because of egress costs. Designed an in-region MSK → S3 / Iceberg / Glue → Spark on EKS maintenance → Athena architecture with snapshot-based watermarks.
Results	Backfilled a representative dataset of ~5 billion rows / 9TB ; audited 196 tables with no missing live rows or unapplied deletes. Repartitioning by delete locality cut large-table MERGE cost per run by ~97% .
Operations	Linked compaction, snapshot expiry and S3 noncurrent-version lifecycle policies, reducing the primary bucket from 259TiB to 120TiB (-54%) and replacing recurring manual work with policies.
Tools	AWS S3, EKS, MSK, Glue, Athena · Iceberg · Spark · Kafka Connect · Airflow · Parquet

Spark on EKS / Karpenter compute and scheduling optimization

08/2026 - Present

Problem	After stopping a cost leak, the executor fleet still cost ~\$830/day. Planned memory utilization was below 45% on 81% of large nodes; the long-job pool lacked Spot capacity.
Analysis	Separated memory, CPU, disk and churn using a 41-application heap census and 775 event logs. Found that 100Gi scratch requests on fixed 512Gi EBS volumes capped every node at four Pods , regardless of instance size.
Decisions	Rejected four alternatives using measurements. Added an NVMe instance-store pool with EBS fallback, reduced executor memory from 32g to 24g, and broadened instance-family fallback for long jobs.
Results	Verified 3.2x allocatable ephemeral storage and removal of the four-Pod cap in production. Long-job launch failures fell from 69-82% to 26% / 2% in observed windows. Quantified ~\$2.1K/month in sparse-node overpayment ; realized savings remain under validation.
Tools	EKS · Karpenter · EC2 Spot · NVMe instance store · EBS · Spark · Terraform · Datadog

Selected Projects: Reliability, Customer Architecture and Enablement

Kafka Connect CDC correctness and recovery

11/2025 - Present

Problem	Source stalls, zombie coordinators and merge-window defects silently lost data even when connectors were RUNNING and consumer lag was zero.
Decisions	Defined end-to-end liveness through commit age, freshness and source-target parity . Applied snapshot watermarks, idempotent anti-join backfills and an Iceberg sink runtime upgrade.
Results	Recovered quantified losses of up to ~355K rows , verified through exact reconciliation. Validated CDC ingestion of 12M+ consent documents for regulatory requirements with zero duplicates and zero dead-letter queue (DLQ) entries.
Practice	Stopped treating lag=0 or RUNNING status as proof of health. Required natural-key and anti-join idempotency for at-least-once replay.
Tools	Amazon MSK · Strimzi Kafka Connect · MongoDB CDC · Iceberg · BigQuery · Airflow · Datadog

Event-ingestion loss protection

03/2026 - 07/2026

Problem	Broker or transport failures could lose producer events; the legacy ingestion endpoint had no retry protection.
Decisions	Implemented failure classification, an S3 write-ahead log and Step Functions / SQS replay from the CTO's fallback PRD. Replaced managed transport with synchronous S3 micro-batches after finding a cross-account Pod Identity defect.
Results	Delivered across four environments through 18 PRs, 95 tests and end-to-end validation. Verified one real production auto-recovery with zero loss and zero duplicates .
Tools	S3 · Step Functions · SQS · EKS Pod Identity · Python · Kubernetes · Terraform · CI/CD

Client on-premises-to-AWS modernization PoC

08/2025 - 11/2025

Problem	Test whether repeated manual setup of client YARN / Spark environments could become reusable cloud infrastructure with least-privilege access.
Decisions	Implemented ~70 AWS / Kubernetes resources end to end: EMR on EKS with Spot scaling during batch windows, per-Pod IRSA, private EKS and Terraform.
Results	Verified reproducibility with destroy-and-reapply deployment and modeled ~46% potential savings . A PoC projection, not measured billing savings; the project ended before production adoption.
Tools	EMR on EKS · S3 · IAM / IRSA · EC2 Spot · Kubernetes · Terraform · Airflow · Helm

Multiregion governance IaC and self-service

01/2026 - Present

Problem	IAM, views, column classification and sensitive-data access were fragmented across 13 projects, ~314 datasets and ~100 users.
Decisions	Created the governance IaC repository, bringing a dataset factory, multiregion layout, keyless CI/CD and tier / IAM model into one Terraform source of truth .
Results	Deployed a new region in ~1 hour including troubleshooting, versus several days estimated for manual setup. All four data engineers use the repository; another engineer independently extended its grant pattern.
Tools	Terraform · GitHub Actions · BigQuery · IAM · Pub/Sub · Python · GitOps

Technical Skills

AWS & Cloud Platform

S3, EKS, MSK, Glue, Athena, EMR on EKS, IAM / IRSA, Step Functions, SQS, EC2 Spot, EBS, Terraform, Kubernetes, Helm, ArgoCD

Data & Reliability

Apache Iceberg, Apache Spark, Kafka Connect, Airflow, BigQuery, Snowflake, Parquet, dbt, Datadog, Carpenter, CDC, Python, SQL

Education and Additional Information

Education	Busan University of Foreign Studies Paideia Creative Talent Studies 02/2012 - 02/2019 · GPA 3.76/4.5
English	TOEIC 930 · 02/2021

Open source	Apache Iceberg PR #17504 Proposal: sort-order overlap statistics
Case studies	Portfolio · English